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Cooling system for a work vehicle

Abstract

A cooling system for a work vehicle includes a fan rotating in a first direction to generate an air flow in a first direction, a speed sensor sensing a speed of the fan, one or more sensors sensing a load of a power source, a temperature sensor sensing an ambient temperature, and a controller configured to determine the load of the power source, determine the speed of the fan, determine an estimated speed of the fan based in part on the load of the power source and the ambient temperature, determine a speed difference of the speed of the fan and the estimated speed of the fan, and determine changing the air flow to a second direction based in part on the speed difference and the load of the power source.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates to a cooling system of a work vehicle.

BACKGROUND

(2) Work vehicles can include cooling fans for circulating air over a radiator during normal operation of the machine. The cooling fan and radiator can be housed in an engine compartment that includes openings for air to flow through the compartment. Typically, these openings are covered with a grill or screen to limit debris from entering the engine compartment.

SUMMARY

(3) According to an aspect of the present disclosure, a cooling system for a work vehicle includes a

fan rotating in a first direction to generate an air flow in a first direction, a speed sensor sensing a speed of the fan, one or more sensors sensing a load of a power source, a temperature sensor sensing an ambient temperature, and a controller configured to determine the load of the power source, determine the speed of the fan, determine an estimated speed of the fan based in part on the load of the power source and the ambient temperature, determine a speed difference of the speed of the fan and the estimated speed of the fan, and determine changing the air flow to a second direction based in part on the speed difference and the load of the power source.

(4) According to an aspect of the present disclosure, the controller is configured to change the air flow to the second direction for a period of time.

(5) According to an aspect of the present disclosure, the controller is configured to change the air flow back to the first direction after the period of time.

(6) According to an aspect of the present disclosure, the controller determines the period of time based in part on the speed difference and the load of the power source.

(7) According to an aspect of the present disclosure, the period of time is a first preselected amount of time.

(8) According to an aspect of the present disclosure, the controller is configured to provide notification when the number of changes in the air flow for a second preselected amount of time reaches or exceeds a rate threshold.

(9) According to an aspect of the present disclosure, the controller is configured to change the air flow to the second direction when the load of the power source is at or below a load threshold and the speed difference is at or above a speed threshold.

(10) According to an aspect of the present disclosure, the controller is configured to change the air flow to the second direction by reversing the rotation of the fan.

(11) According to an aspect of the present disclosure, the controller is configured to change the air flow to the second direction by adjusting a pitch of a fan blade of the fan.

(12) According to an aspect of the present disclosure, the controller is configured to provide notification when the speed difference is at or above a threshold.

(13) The above and other features will become apparent from the following detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description refers to the accompanying figures.

(2) FIG. 1 is a perspective view of a work vehicle including a cooling system, according to an implementation.

(3) FIG. 2 is a schematic diagram of a cooling system for a work vehicle, according to an implementation.

(4) FIG. 3 is a map of a field, according to an implementation.

(5) FIG. 4 is a schematic diagram of a cooling system for a work vehicle, according to an implementation.

(6) FIG. 5 is a flow diagram of a cooling system for a work vehicle, according to an implementation.

(7) Like reference numerals are used to indicate like elements throughout the several figures.

DETAILED DESCRIPTION

(8) The embodiments or implementations disclosed in the above drawings and the following detailed description are not intended to be exhaustive or to limit the present disclosure to these embodiments or implementations.

(9) With reference to FIG. 1, a work vehicle **100**, for example an agricultural tractor, can include an

operator station or cab **102**, a hood **104**, one or more ground engaging apparatus **106**, for example wheels or track assemblies, and a frame or chassis **110**. The work vehicle **100** can include one or more power sources **108**, for example an internal combustion engine, a hybrid engine, and a battery and an electric machine. The work vehicle **100** can include an operator interface having any number and combination of electronic devices, such as an interactive display. The work vehicle **100** can have a rigid or an articulated frame **110**. The work vehicle **100** can include a cooling system **120** for removing heat from various components and systems during operation. This disclosure also applies to other types of work vehicles in agriculture, construction, forestry, and road building.

(10) With reference to FIG. 2, a cooling system **120** for a work vehicle **100** can include a fan **122** positioned within a compartment **112** at least partially defined by the hood **104**. The fan **122** can be positioned near a grill or screen **114** and a cooling element **116**, for example a radiator. The fan **122** includes one or more fan blades **124** and a fan motor **126**. The fan **122** can rotate in a first direction creating an air flow in a first direction **130** from the screen **114** towards the cooling element **116**. The fan **122** can rotate in a second direction creating an air flow in a second direction **132**, which is opposite of the first direction, from the cooling element **116** towards the screen **114**. Alternatively, or additionally, the fan **122** can rotate in one direction and the pitch or angle of the fan blades **124** can be changed to the reverse the direction of the air flow between the first direction **130** and the second direction **132**.

(11) When the fan **122** is rotating in the first direction, material can collect on the screen **114**, which can reduce the cooling capacity of the cooling system. Reversing the direction of the air flow produced by the fan **122** can remove the material from the screen **114**. When the fan **122** is operating in the second direction **132**, the fan **122** can create additional noise and the cooling capacity of the cooling system **120** can be reduced. According to some implementations, the cooling system **120** can reduce or minimize the number of times the direction of the air flow is changed or reversed by determining when a reversal of the air flow is needed.

(12) With reference to FIG. 3, an electronic map **140** of a field **142** includes a field boundary **144** around the perimeter of the field. The map **140** can include a proposed or actual route **150** for a work vehicle **100** to travel through the field **142** and perform one or more operations, for example planting, tilling, spraying, mowing, harvesting, etc. The map **140** can include one or more areas **152** of an anticipated or predicted higher load on the power source **108** (e.g., sloped or uphill terrain, low-traction soil conditions, etc.).

(13) With reference to FIGS. 1-4, an electronic control unit or controller **160** can connect to and communicate with the electronic map **140**, which can be stored locally or remotely, a Global Positioning System (GPS) receiver **162**, a speed sensor **128** detecting or measuring the rotational speed of the fan **122**, a temperature sensor **164** detecting or measuring an ambient temperature, a temperature sensor **138** detecting or measuring a temperature of a coolant in the cooling element **116**, and one or more sensors **170** detecting or measuring various operations of the work vehicle **100** (e.g., engine sensors, transmission sensors, hydraulic sensors, electrical sensors, speed sensors, pressure sensors, temperature sensors, torque sensors, fuel flow sensors, air flow sensors, etc.). The controller **160** can connect to and communicate with a pneumatic or hydraulic system **166** or an electrical system **168** of the work vehicle **100** to change or reverse the direction of the air flow of the fan **122**. For example, the controller **160** can connect to and communicate with a valve or actuator of the pneumatic or hydraulic system **166** to set the rotational speed and direction of the fan **122** and to change or reverse the rotational direction of the fan **122** or change the pitch or angle of the fan blades **124**. Alternatively, or additionally, the controller **160** can connect to and communicate with a switch, relay, actuator, or controller of the electrical system **168** to set the rotational speed and direction of the fan **122** and to change or reverse the rotational direction of the fan **122** or change the pitch or angle of the fan blades **124**.

(14) The controller **160** can determine the load of the power source **108**. The controller **160** can communicate with a power source controller **118** (e.g., engine controller, battery management

system, etc.) to determine the load of the power source **108**. The controller **160** can communicate with the one or more sensors **170** (e.g., air flow sensor, fuel flow sensor, engine speed sensor, etc.) to determine the load of the power source **108**. The controller **160** can determine whether the load of the power source **108** is at a high or low load or anywhere in between. The controller **160** can determine whether the load of the power source **108** is below, at, or above a load threshold, which can be a percentage of a maximum load of the power source **108**. The load threshold can be preset, preselected, or determined via the controller **160**.

(15) The controller **160** can determine the actual speed of the fan **122** based on the set rotational speed of the fan **122** or detecting or measuring the speed of the fan **122** via a speed sensor **128**. The controller **160** can determine an estimated speed of the fan **122** based on one or more of the load of the power source **108**, the ambient temperature detected or measured via the temperature sensor **164**, and the coolant temperature detected or measured via the temperature sensor **138**. The estimated speed of the fan **122** increases as the load of the power source **108** increases. The estimated speed of the fan **122** increases as the ambient temperature increases. The estimated speed of the fan **122** increases as the coolant temperature increases. The load of the power source **108**, the ambient temperature, and the coolant temperature can influence the estimated speed of the fan **122** to the same or a different extent. For example, the load of the power source **108** can have more of an effect on the estimated speed of the fan **122** than the ambient temperature or the coolant temperature, the ambient temperature can have more of an effect on the estimated speed of the fan **122** than the load of the power source **108** or the coolant temperature, or the coolant temperature can have more of an effect on the estimated speed of the fan **122** than the load of the power source **108** or the ambient temperature. The controller **160** can determine a speed difference between the actual speed of the fan **122** and the estimated speed of the fan **122**. The controller **160** can determine whether the speed difference is below, at, or above a speed threshold, which can be a percentage of the estimated speed or actual speed of the fan **122**. The speed threshold can be preset, preselected, or determined via the controller **160**.

(16) The controller **160** can determine whether to change or reverse the air flow from a first direction **130** to a second direction **132** based in part on the speed difference, the load of the power source **108**, or both. The controller **160** can determine to change or reverse the air flow from the first direction **130** to the second direction **132** when the speed difference is at or above the speed threshold and the load of the power source **108** is at or below the load threshold. The controller **160** can change the air flow to the second direction **132** for a period of time. The controller **160** can change the air flow back to the first direction **130** after the period of time. The period of time can be preset or preselected. Alternatively, or additionally, the controller **160** can determine the period of time based in part on one or more of the speed difference, the load of the power source **108**, the ambient temperature, and the coolant temperature. For an example, a larger the speed difference can result in a longer the period of time. For another example, the period of time can continue until the speed difference reduces to or below the speed threshold, the load of the power source **108** increases to or above the load threshold, or both. The controller **160** can change the air flow to the second direction **132** by reversing the rotation of the fan **122**. Alternatively, or additionally, the controller **160** can change the air flow to the second direction **132** by adjusting a pitch of the fan blade **124**.

(17) The controller **160** can determine the number of changes or reversals of the air flow in a preset or preselected amount of time. The controller **160** can provide an audio or visual notification, or both, via a local, remote, or mobile electronic device **180** (e.g., vehicle display, remote computer, cell phone, etc.) when the number of changes or reversals of the air flow in the preselected amount of time reaches or exceeds a rate threshold, which can be preset, preselected, or determined via the controller **160**. The controller **160** can provide an audio or visual notification via the electronic device **180** when the speed difference is below, at, or above the speed threshold. The controller **160** can provide an audio or visual notification via the electronic device **180** when the load of the power

source **108** is below, at, or above the load threshold. The controller **160** can provide an audio or visual notification via the electronic device **180** when the number of changes or reversals of the air flow in the preselected amount of time reaches or exceeds the rate threshold.

(18) With reference to FIG. 5, a cooling system **120** for a work vehicle **100** includes one or more of the following steps, processes, or operations. At **200**, the controller **160** monitors the cooling system **120**. At **202**, the controller **160** determines the ambient temperature via the temperature sensor **164**. At **204**, the controller **160** determines the coolant temperature via the temperature sensor **138**. At **206**, the controller **160** determines the load of the power source **108** via the power source controller **118**, the one or more sensors **170**, or both. At **208**, the controller **160** determines the actual fan speed by the set rotational speed of the fan **122**, the speed sensor **128**, or both. At **210**, the controller **160** determines the estimated fan speed via the load of the power source **108**, the ambient temperature, and the coolant temperature.

(19) At **212**, the controller **160** determines the speed difference between the actual fan speed and the estimated fan speed. At **214**, the controller **160** determines whether the load of the power source is at or below a load threshold. At **216**, the controller **160** determines whether the speed difference is at or above a speed threshold. If either **214** or **216** are false, then the controller **160** continues monitoring the cooling system **120** at **200**. If both **214** and **216** are true, then at **218** the controller **160** changes the air flow from a first direction **130** to a second direction **132** for a period of time, and then changes the air flow back to the first direction **130**. The controller **160** then continues monitoring the cooling system **120** at **200**. In other implementations, one or more of these steps, processes, or operations may be omitted, repeated, re-ordered, combined, or separated and are within the scope of the present disclosure.

(20) The electronic control unit(s) or controller(s) disclosed herein can have one or more microprocessor-based electronic control units or controllers, which perform calculations and comparisons and execute instructions, for example algorithms. The controller includes a processor, a core, volatile and non-volatile memory, digital and analog inputs, and digital and analog outputs. The controller can include non-transitory, computer readable memory, such as random-access memory (RAM), read only memory (ROM), or electrically erasable programmable read only memory (EEPROM), which include instructions for execution by the processor, for example algorithms. The controller connects to and communicates with various input and output devices including, but not limited to, switches, relays, solenoids, actuators, light emitting diodes (LED's), passive and interactive displays, radio frequency devices (RFD's), sensors, and other controllers. The controller receives communications or signals, via electrically or any suitable electromagnetic communication, from one or more devices, determines an appropriate response or action, and sends communications or signals to one or more devices. The controller can be a microprocessor, an application specific integrated circuit (ASIC), a digital processor, or a programmable logic controller, also known as a PLC or programmable controller. The controller can connect to and communicate with an electronic control system of the work vehicle through a data bus, such as a CAN bus, or the controller can be a part the electronic control system of the work vehicle.

(21) The terminology used herein is for the purpose of describing example embodiments or implementations and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the any use of the terms “has,” “includes,” “comprises,” or the like, in this specification, identifies the presence of stated features, integers, steps, operations, elements, and/or components, but does not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(22) Those having ordinary skill in the art will recognize that terms such as “above,” “below,” “upward,” “downward,” “top,” “bottom,” etc., are used descriptively for the figures, and do not represent limitations on the scope of the present disclosure, as defined by the appended claims.

Furthermore, the teachings may be described herein in terms of functional and/or logical block components or various processing steps, which may include any number of hardware, software, and/or firmware components configured to perform the specified functions.

(23) Terms of degree, such as “generally,” “substantially,” or “approximately” are understood by those having ordinary skill in the art to refer to reasonable ranges outside of a given value or orientation, for example, general tolerances or positional relationships associated with manufacturing, assembly, and use of the described embodiments or implementations.

(24) As used herein, “e.g.,” is utilized to non-exhaustively list examples and carries the same meaning as alternative illustrative phrases such as “including,” “including, but not limited to,” and “including without limitation.” Unless otherwise limited or modified, lists with elements that are separated by conjunctive terms (e.g., “and”) and that are also preceded by the phrase “one or more of” or “at least one of” indicate configurations or arrangements that potentially include individual elements of the list, or any combination thereof. For example, “at least one of A, B, and C” or “one or more of A, B, and C” indicates the possibilities of only A, only B, only C, or any combination of two or more of A, B, and C (e.g., A and B; B and C; A and C; or A, B, and C).

(25) While the above describes example embodiments or implementations of the present disclosure, these descriptions should not be viewed in a restrictive or limiting sense. Rather, there are several variations and modifications which may be made without departing from the scope of the appended claims.

Claims

1. A cooling system for a work vehicle, comprising: a fan rotating in a first direction to generate an air flow in a first direction; a speed sensor sensing a speed of the fan; one or more sensors sensing a load of a power source; a temperature sensor sensing an ambient temperature; and a controller configured to determine the load of the power source, determine the speed of the fan, determine an estimated speed of the fan based in part on the load of the power source and the ambient temperature, determine a speed difference of the speed of the fan and the estimated speed of the fan, and determine changing the air flow to a second direction based in part on the speed difference and the load of the power source.
2. The cooling system of claim 1, wherein the controller is configured to change the air flow to the second direction for a period of time.
3. The cooling system of claim 2, wherein the controller is configured to change the air flow back to the first direction after the period of time.
4. The cooling system of claim 2, wherein the controller determines the period of time based in part on the speed difference and the load of the power source.
5. The cooling system of claim 2, wherein the period of time is a first preselected amount of time.
6. The cooling system of claim 3, wherein the controller is configured to provide notification when a number of changes in the air flow for a second preselected amount of time reaches or exceeds a rate threshold.
7. The cooling system of claim 1, wherein the controller is configured to change the air flow to the second direction when the load of the power source is at or below a load threshold and the speed difference is at or above a speed threshold.
8. The cooling system of claim 1, wherein the controller is configured to change the air flow to the second direction by reversing the rotation of the fan.
9. The cooling system of claim 1, wherein the controller is configured to change the air flow to the second direction by adjusting a pitch of a fan blade of the fan.
10. The cooling system of claim 1, wherein the controller is configured to provide notification when the speed difference is at or above a threshold.
11. A work vehicle, comprising: one or more ground engaging apparatus; a power source; a fan

rotating in a first direction to generate an air flow in a first direction; a speed sensor sensing a speed of the fan; one or more sensors sensing a load of a power source; a temperature sensor sensing an ambient temperature; and a controller configured to determine the load of the power source, determine the speed of the fan, determine an estimated speed of the fan based in part on the load of the power source and the ambient temperature, determine a speed difference of the speed of the fan and the estimated speed of the fan, and determine changing the air flow to a second direction based in part on the speed difference and the load of the power source.

12. The work vehicle of claim 11, wherein the controller is configured to change the air flow to the second direction for a period of time.

13. The work vehicle of claim 12, wherein the controller is configured to change the air flow back to the first direction after the period of time.

14. The work vehicle of claim 12, wherein the controller determines the period of time based in part on the speed difference and the load of the power source.

15. The work vehicle of claim 12, wherein the period of time is a first preselected amount of time.

16. The work vehicle of claim 13, wherein the controller is configured to provide notification when a number of changes in the air flow for a second preselected amount of time reaches or exceeds a rate threshold.

17. The work vehicle of claim 11, wherein the controller is configured to change the air flow to the second direction when the load of the power source is at or below a load threshold and the speed difference is at or above a speed threshold.

18. The work vehicle of claim 11, wherein the controller is configured to change the air flow to the second direction by reversing the rotation of the fan.

19. The work vehicle of claim 11, wherein the controller is configured to change the air flow to the second direction by adjusting a pitch of a fan blade of the fan.

20. The work vehicle of claim 11, wherein the controller is configured to provide notification when the speed difference is at or above a threshold.
